

Introduction to Numerical Algorithms

Class 3.

Solution of system of linear equations

System of linear equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

$(x_1, x_2, \dots, x_n)$ are the unknowns $\rightarrow$ can be collected to a vector $\underline{x}$

a_{ij} are the coefficients of the equations $\rightarrow$ can be collected to a matrix $\underline{A}$

$(b_1, b_2, \dots, b_n)$ are constant terms $\rightarrow$ can be collected to a vector $\underline{b}$

$$\underline{A} \cdot \underline{x} = \underline{b} \quad \text{in short}$$

Goal: for given $\underline{A}$ and $\underline{b}$ what is $\underline{x}$?

for that we construct the augmented matrix $(\underline{A} | \underline{b})$

and try to make (i) the values below the diagonal $\emptyset$
(ii) the diagonal 1 $\rightarrow$ optional

How to solve this puzzle?

Legal moves:

- Swapping two rows $\rightarrow$ equation rearrangement
- Scaling rows $\rightarrow$ multiply an equation with a non-zero number
- Add multiple of one row to another row $\rightarrow$ combining equations

How it works in practice?

②

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$x + 2y - z = 2$$

$$\Rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix}$$

(1) Eliminate everything below a_{11}

$$\begin{pmatrix} 1 & 1 & 1 & | & 6 \\ 0 & -3 & -1 & | & -9 \\ 0 & 1 & -2 & | & -4 \end{pmatrix} \begin{array}{l} \\ R_2 - 2 \cdot R_1 \\ R_3 - 1 \cdot R_1 \end{array}$$

(2) Scale a_{22} to 1, eliminate everything below

$$\begin{pmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 1/3 & | & 3 \\ 0 & 1 & -2 & | & -4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 1/3 & | & 3 \\ 0 & 0 & -7/3 & | & -7 \end{pmatrix}$$

(3) Scale a_{33} to 1, (4.) Back substitution

$$\begin{pmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 1/3 & | & 3 \\ 0 & 0 & 1 & | & 3 \end{pmatrix}$$

$$z = 3$$

$$y = 3 - 1/3 \cdot z = 2$$

$$x = 6 - y - z = 1$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

The pseudocode for the GE algorithm

for $i = 1 \dots n-1$ #iterate columns from 1 to one before last
find in column i a non-zero element between
rows $i \dots n$, swap if necessary

for $j = i \dots n$

by subtracting the multiples of row i from row j

kill $A[j, i]$

③

GE gives upper triangle matrix, how to reconstruct $\underline{x}$?

Backward substitution

we did it manually before

x_i can be reconstructed from $x_{i+1} \dots x_n$
in a recursive way

the result of GE is:

$$\underline{U} \cdot \underline{x} = \underline{b}'$$

from that x_i can be expressed

$$x_i = b'_i - u_{i,i+1} x_{i+1} - u_{i,i+2} x_{i+2} - \dots - u_{i,n} x_n$$

or in vector notation if $\underline{x} = \underline{0}$ in the beginning

$$x_i = b'_i - \underline{u}_i \cdot \underline{x}$$

where $\underline{u}_i$ is the i th row of $\underline{U}$
start from row n and go to row 1

Why backward substitution?

Computational cost:

GE (to upper triangle): $\mathcal{O}(\frac{2}{3}n^3)$ Δ almost 2x faster
backward substitution: $\mathcal{O}(n^2)$

GE (to diagonal) $\mathcal{O}(\frac{4}{3}n^3)$

NOTE: we solve $\underline{A} \underline{x} = \underline{b}$ without calculating $\underline{A}^{-1}$
and much faster

What can go wrong?

④

(a) division by zero

$$0 \cdot x + 2y = 4$$

$$1 \cdot x + 3 \cdot y = 7$$

we need to swap rows (equations) to avoid zero division

(b) instability

$$10^{-5}x + y = 1$$

$$x + y = 2$$

using 10^{-5} for elimination can increase numerical errors
large multiplier, amplifies errors.
We can always change equations to smartly avoid large errors!

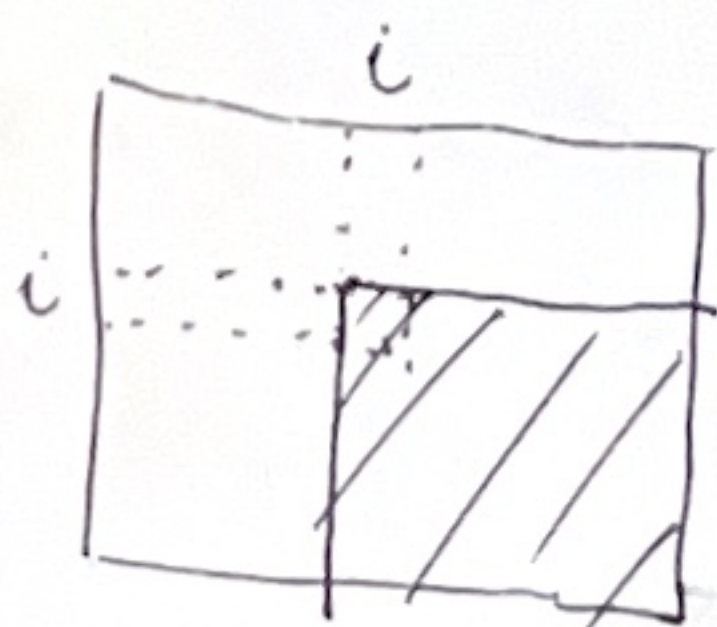
(1) Partial pivoting

- only change rows (equations)
- use the largest element from column i starting from row i to eliminate



(2) Full pivoting

- can change both rows (equations) and columns (variables)
- use the largest element from the remaining submatrix $A[i:n, i:n]$, bring this element by row and column swaps to $A[i, i]$



What happens if we want to solve many r.h.s.?

5

$$\underline{A} \underline{x} = \underline{b}_n \quad \text{with different } \underline{b}_n \text{'s}$$

we can use the previous Gauss elimination if we store the elimination factors in $\underline{L}$

$$\underline{L} \cdot \underline{U} = \underline{A}$$

Gauss elimination with memory

$$\underline{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{pmatrix}$$

$$\underline{L} = \begin{pmatrix} 1 & 0 & 0 \\ ? & 1 & 0 \\ ? & ? & 1 \end{pmatrix}$$

$$\underline{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -3 & -1 \\ 0 & 1 & -2 \end{pmatrix}$$

$$\underline{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & ? & 1 \end{pmatrix}$$

$$\underline{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -3 & -1 \\ 0 & 0 & -7/3 \end{pmatrix}$$

$$\underline{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -1/3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -3 & -1 \\ 0 & 0 & -7/3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{pmatrix} \quad \checkmark$$

The solution:

$$\underline{L} \cdot \underline{U} \cdot \underline{x} = \underline{b} \quad \text{new variable } \underline{y} = \underline{U} \cdot \underline{x}$$

(1) $\underline{L} \cdot \underline{y} = \underline{b}$ with forward substitution $\rightarrow$ we have $\underline{y}$

(2) $\underline{U} \cdot \underline{x} = \underline{y}$ with backward substitution

$\rightarrow$ cheaper than doing GE many times

⑥

Pivoting and LU

(1) Partial pivoting

- track the swaps in $\underline{P}$

$$\underline{P} \underline{A} = \underline{L} \cdot \underline{U} \quad \text{then solve} \quad \underline{L} \underline{U} = \underline{P} \cdot \underline{b}$$

(2) Full pivoting

- track row swaps in $\underline{P}$

- track column swaps in $\underline{Q}$

$$\underline{P} \underline{A} \underline{Q} = \underline{L} \underline{U} \quad \text{then solve} \quad \underline{L} \underline{U} = \underline{P} \underline{b} \underline{Q}$$